

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2**

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**COURSE NAME: COMPUTER NETWORKS**

**COURSE CODE: CSA0714**

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### **COURSE OUTCOME COVERED**

**CO4:**

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

*Assessment Tool Weightage: 30%*

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Annexure A

### **Industry Problem-Solving Task**

1. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

## **IMPLEMENTATION IN VS CODE:**

### **APP.PY:**

```
from flask import Flask, render_template, jsonify

app = Flask(__name__)

departments = {

    "HR": {

        "vlan": 10,

        "network": "192.168.10.0/24",

        "gateway": "192.168.10.1",

        "devices": ["HR-PC1", "HR-PC2", "HR-Printer"]

    },

    "Finance": {

        "vlan": 20,

        "network": "192.168.20.0/24",

        "gateway": "192.168.20.1",

        "devices": ["Finance-PC1", "Finance-PC2", "Finance-Printer"]

    },

    "Sales": {

        "vlan": 30,

        "network": "192.168.30.0/24",

        "gateway": "192.168.30.1",
```

```
    "evices": ["Sales-PC1", "Sales-PC2", "Sales-Printer"]

},

"IT": {

    "vlan": 40,

    "network": "192.168.40.0/24",

    "gateway": "192.168.40.1",

    "devices": ["IT-PC1", "IT-PC2", "IT-Server"]

}
```

```
@app.route("/")
```

```
def home():
```

```
    return render_template("index.html", departments=departments)
```

```
@app.route("/network-status")
```

```
def network_status():
```

```
    return jsonify({
```

```
        "status": "Network Online",
```

```
        "layer2": "Access Switches Connected",
```

```
        "layer3": "Inter-VLAN Routing Active",
```

```
        "internet": "Internet Gateway Connected"
```

```
    })
```

```
@app.route("/ping/<source>/<destination>")
```

```
def ping(source, destination):
```

```
if source not in departments or destination not in departments:
```

```
    return jsonify({
        "success": False,
        "message": "Invalid department"
    })
```

```
if source == destination:
```

```
    return jsonify({
        "success": True,
        "message": f"{source} devices can communicate directly within the same VLAN."
    })
```

```
return jsonify({
```

```
    "success": True,
    "message": (
        f"{source} → Layer 2 Switch → "
        f"Layer 3 Switch → Inter-VLAN Routing → "
        f"{destination} Switch → {destination}"
    )
})
```

```
if __name__ == "__main__":
```

```
    app.run(debug=True)
```

## INDEX.HTML

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Company Network Simulator</title>
```

```
  <link rel="stylesheet" href="/static/style.css">
```

```
</head>
```

```
<body>
```

```
<h1>Company Network Design & VLAN Simulator</h1>
```

```
<p class="subtitle">
```

```
  Mini implementation of a hierarchical company network
```

```
</p>
```

```
<div class="topology">
```

```
  <div class="internet">
```

```
     INTERNET
```

```
  </div>
```

```
  <div class="router">
```

```
     ROUTER / FIREWALL
```

```
    <small>192.168.1.1</small>
```

```
  </div>
```

```
<div class="connection">↓</div>
```

```
<div class="core">
```

```
   LAYER 3 SWITCH
```

<small>Inter-VLAN Routing</small>

</div>

<div class="connection">↓</div>

<div class="departments"

{% for name, data in departments.items() %}

<div class="department">

<h2>{{ name }}</h2>

<div class="switch">

 L2 SWITCH

</div>

<p>

<b>VLAN:</b> {{ data.vlan }}

</p>

<p>

<b>Network:</b> {{ data.network }}

</p>

<p>

<b>Gateway:</b> {{ data.gateway }}

</p>

<div class="devices">

{% for device in data.devices %}

<div class="device">

 {{ device }}

```
        </iv>

        {% endfor %}

    </div>

</div>

{% endfor %}

</div>

</div>

<div class="control-panel">

    <h2>Network Testing</h2>

    <label>Source Department:</label>

    <select id="source">

        {% for name in departments %}

            <option value="{{ name }}">{{ name }}</option>

        {% endfor %}

    </select>

    <label>Destination Department:</label>

    <select id="destination">

        {% for name in departments %}

            <option value="{{ name }}">{{ name }}</option>

        {% endfor %}

    </select>

    <button onclick="testConnection()">

        Test Communication

    </button>

</div>
```

</button>

<div id="result"></div>

</div>

<div class="info"

<h2>Network Configuration</h2>

<table>

<tr>

<th>Department</th>

<th>VLAN</th>

<th>Network</th>

<th>Gateway</th>

</tr>

{% for name, data in departments.items() %

<tr>

<td>{{ name }}</td>

<td>{{ data.vlan }}</td>

<td>{{ data.network }}</td>

<td>{{ data.gateway }}</td>

</tr>

{% endfor %}

</table>

</div>



```
<script>

function testConnection() {

    let source = document.getElementById("source").value;

    let destination = document.getElementById("destination").value;

    fetch(`/ping/${source}/${destination}`)

        .then(response => response.json())

        .then(data => {

            let result = document.getElementById("result");

            result.innerHTML = `

                <div class="success">

                     ${data.message}

                </div>

            `;

        });

    }

</script>

</body>

</html>
```

## STYLE.CSS

```
body {

    font-family: Arial, sans-serif;

    background: #f2f5f9;

    margin: 0;
```

```
padding: 20px;

text-align: center;

}
```

```
h1 {

color: #1f2937;

}
```

```
.subtitle {

color: #555;

}
```

```
.topology {

margin: 30px auto;

max-width: 1200px;

}
```

```
.internet {

background: #2563eb;

color: white;

padding: 15px;

width: 180px;

margin: auto;
```

```
border-radius: 10px;  
  
font-weight: bold;  
  
}
```

```
.router {  
  
background: #dc2626;  
  
color: white;  
  
padding: 15px;  
  
width: 220px;  
  
margin: 20px auto;  
  
border-radius: 10px;  
  
font-weight: bold;  
  
}
```

```
.router small,  
  
.core small {  
  
display: block;  
  
margin-top: 5px;  
  
}
```

```
.connection {  
  
font-size: 30px;  
  
}
```

```
.core {  
  
    background: #7c3aed;  
  
    color: white;  
  
    padding: 20px;  
  
    width: 260px;  
  
    margin: auto;  
  
    border-radius: 10px;  
  
    font-weight: bold;  
  
}
```

```
.departments {  
  
    display: grid;  
  
    grid-template-columns: repeat(4, 1fr);  
  
    gap: 15px;  
  
    margin-top: 30px;  
  
}
```

```
.department {  
  
    background: white;  
  
    padding: 15px;  
  
    border-radius: 12px;  
  
    box-shadow: 0 3px 10px #bbb;
```

```
}
```

```
.department h2 {  
  
    margin-top: 0;  
  
}
```

```
.switch {  
  
    background: #0891b2;  
  
    color: white;  
  
    padding: 12px;  
  
    border-radius: 8px;  
  
    margin-bottom: 10px;  
  
}
```

```
.device {  
  
    background: #e5e7eb;  
  
    margin: 6px;  
  
    padding: 8px;  
  
    border-radius: 5px;  
  
}
```

```
.control-panel {  
  
    background: white;  
  
    max-width: 700px;
```

```
margin: 30px auto;
```

```
padding: 25px;
```

```
border-radius: 12px;
```

```
box-shadow: 0 3px 10px #bbb;
```

```
select,
```

```
button {
```

```
padding: 10px;
```

```
margin: 8px;
```

```
}
```

```
button {
```

```
background: #2563eb;
```

```
color: white;
```

```
border: none;
```

```
border-radius: 6px;
```

```
cursor: pointer;
```

```
}
```

```
button:hover {
```

```
background: #1d4ed8;
```

```
}
```

```
.success {
```

```
margin-top: 20px;
```

```
background: #dcfce7;

padding: 15px;

border-radius: 8px;

color: #166534;

}

.info {

    max-width: 900px;

    margin: 30px auto;

    background: white;

    padding: 20px;

    border-radius: 12px;

}

table {

    width: 100%;

    border-collapse: collapse;

}

th,

td {

    border: 1px solid #ccc;

    padding: 10px;

}

th {
```

```
background: #e5e7eb;  
  
}
```

```
@media (max-width: 900px) {
```

```
  .departments {  
  
    grid-template-columns: repeat(2, 1fr);  
  
  }
```

```
}
```

```
@media (max-width: 600px) {
```

```
  .departments {  
  
    grid-template-columns: 1fr;  
  
  }
```

```
}
```



OUTPUT:

# Company Network Design & VLAN Simulator

Mini implementation of a hierarchical company network

INTERNET

ROUTER / FIREWALL  
192.168.1.1

LAYER 3 SWITCH  
Inter-VLAN Routing

HR

L2 SWITCH

VLAN: 10

Network: 192.168.10.0/24

Gateway: 192.168.10.1

HR-PC1

HR-PC2

HR-Printer

Finance

L2 SWITCH

VLAN: 20

Network: 192.168.20.0/24

Gateway: 192.168.20.1

Finance-PC1

Finance-PC2

Finance-Printer

Sales

L2 SWITCH

VLAN: 30

Network: 192.168.30.0/24

Gateway: 192.168.30.1

Sales-PC1

Sales-PC2

Sales-Printer

IT

L2 SWITCH

VLAN: 40

Network: 192.168.40.0/24

Gateway: 192.168.40.1

IT-PC1

IT-PC2

IT-Server

Network Testing

Source Department: Finance Destination Department: IT

Test Communication

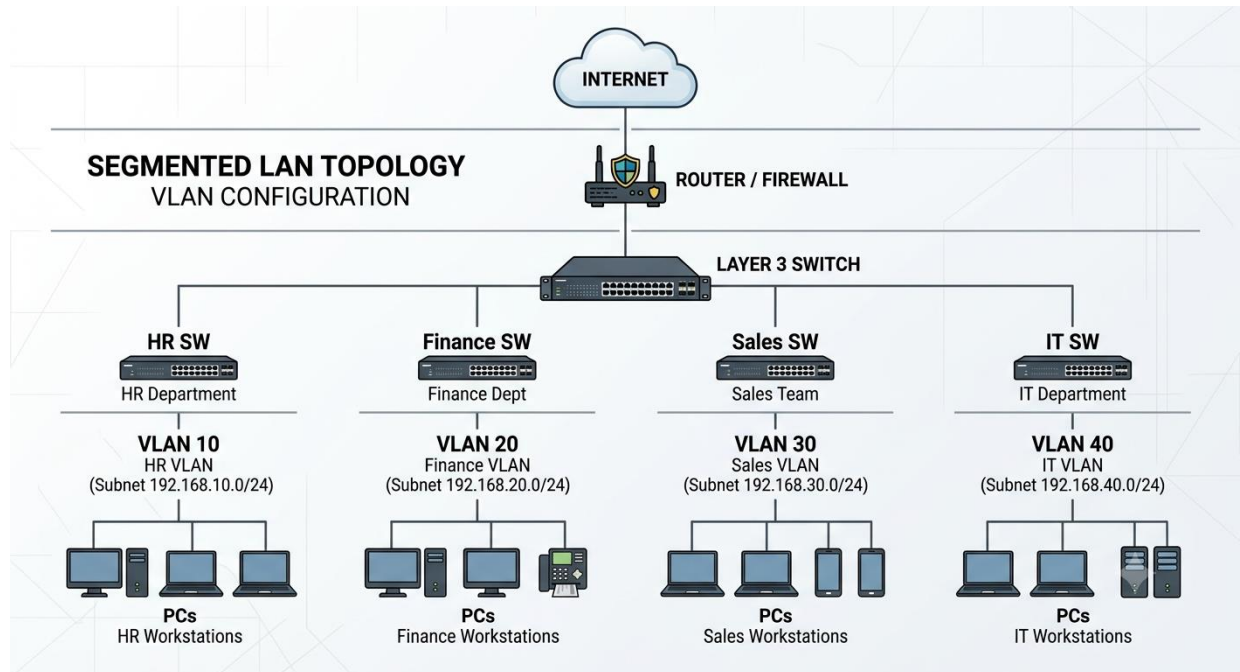
✓ Finance → Layer 2 Switch → Layer 3 Switch → Inter-VLAN Routing → IT Switch → IT

Network Configuration

| Department | VLAN | Network         | Gateway      |
|------------|------|-----------------|--------------|
| HR         | 10   | 192.168.10.0/24 | 192.168.10.1 |
| Finance    | 20   | 192.168.20.0/24 | 192.168.20.1 |
| Sales      | 30   | 192.168.30.0/24 | 192.168.30.1 |
| IT         | 40   | 192.168.40.0/24 | 192.168.40.1 |

## How the project works

Your implementation represents a **3-layer hierarchical network**.



## VLAN configuration

| Department     | VLAN | Network         | Gateway      |
|----------------|------|-----------------|--------------|
| <b>HR</b>      | 10   | 192.168.10.0/24 | 192.168.10.1 |
| <b>Finance</b> | 20   | 192.168.20.0/24 | 192.168.20.1 |
| <b>Sales</b>   | 30   | 192.168.30.0/24 | 192.168.30.1 |
| <b>IT</b>      | 40   | 192.168.40.0/24 | 192.168.40.1 |

## What happens when you test communication?

For example:

**HR → Finance**

HR PC

↓

HR Layer-2 Switch

↓

Layer-3 Switch

↓

Inter-VLAN Routing

↓

Finance Layer-2 Switch

↓

Finance PC

The application displays this when you click **Test Communication**.

For:

**HR → HR**

the application shows that the devices are communicating within the same VLAN.

### **What you can explain during your mini-project presentation**

#### **Problem:**

A company has multiple departments and needs secure and efficient communication.

#### **Solution:**

We divide the network into separate VLANs for HR, Finance, Sales, and IT.

#### **Layer**

**2:**

Each department has an access switch connecting its computers and printers.

#### **Layer3:**

A central Layer 3 switch performs inter-VLAN routing.

#### **Router:**

The router connects the internal company network to the Internet and represents the firewall/security boundary.

**IPaddressing:**

Each department receives its own /24 subnet.

**Security:**

VLAN separation prevents all departments from being part of one large broadcast domain and provides a basis for applying access-control policies.

**Scalability:**

New computers or even new departments can be added without redesigning the entire network.

## RUBRICS FOR CALCULATION

| Criteria                     | Weightage | Exemplary (4)                                                                | Proficient (3)                            | Developing (2)                                | Beginning (1)                                  |
|------------------------------|-----------|------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------|------------------------------------------------|
| Problem Analysis             | 15%       | Clearly identifies and deeply analyzes the industry problem with all factors | Good understanding with minor gaps        | Basic understanding; some key aspects missing | Poor or incorrect understanding of the problem |
| Solution Design              | 20%       | Innovative, practical, and industry-relevant solution                        | Effective solution with minor limitations | Partially suitable solution                   | Inappropriate or unclear solution              |
| Technical Implementation     | 20%       | Fully functional, accurate, and well-executed implementation                 | Mostly functional with minor errors       | Partially implemented solution                | Incomplete or non-functional implementation    |
| Use of Tools & Technologies  | 10%       | Appropriate and advanced use of relevant tools and technologies              | Correct use of tools                      | Limited or basic use of tools                 | Incorrect or no use of tools                   |
| Problem-Solving Approach     | 10%       | Logical, systematic, and well-structured approach                            | Mostly logical approach                   | Somewhat disorganized approach                | No clear approach                              |
| Innovation & Creativity      | 10%       | Highly creative and original solution                                        | Some creativity                           | Limited originality                           | No creativity                                  |
| Testing & Validation         | 5%        | Thorough testing with real-world scenarios                                   | Adequate testing                          | Minimal testing                               | No testing                                     |
| Documentation & Presentation | 5%        | Clear, professional, and well-documented                                     | Good documentation                        | Basic documentation                           | Poor or no documentation                       |
| Teamwork / Collaboration     | 5%        | Excellent coordination and contribution (if team-based)                      | Good collaboration                        | Limited participation                         | No collaboration or contribution               |